

Problema:

Promedio movil toma un vector de tamaño n y un valor k , se van calcular los promedios de cada k valores

$A = \{2, 4, 5, 1, 3\}$
 $k = 3$

$i = 0, 1, 2 \Rightarrow (2+4+5)/3 = 11/3$
 $i = 1, 2, 3 \Rightarrow (4+5+1)/3 = 10/3$
 $i = 2, 3, 4 \Rightarrow (5+1+3)/3 = 9/3 = 3$
arr)
 $A = (x, y, z)$

arr, k

$i = 0, 1, \dots, (n-1) - (k-1)$

$i = 0, 2, \dots, n-k$

```
def promedioMovil(arr: Vector[Int], k: Int): Vector[Int] = {  
  (0 to arr.length-k)  
    .map(i => (0 until k).map(t => arr(i+t)/k.toDouble).sum)
```

$i = 0, 1, 2, \dots, n-k$
 $T1 = 0, 1, 2, \lfloor \frac{n-k}{2} \rfloor$
 $T2 = \lfloor \frac{n-k}{2} \rfloor + 1, \dots, n-k$

$[0, \frac{n-k}{2}]$
 $[\frac{n-k}{2} + 1, n-k]$

$T1 [0, \frac{n-k}{2} + k - 1]$

$T2 [\frac{n-k}{2} + 1, n-1]$

ini, fin

```
scala> arr.slice(0, (arr.length-k)/2 + k)  
val res35: scala.collection.immutable.Vector[Int] = Vector(2, 4, 5, 1)
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scala> arr.slice((arr.length-k)/2 + 1, arr.length)
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val res36: scala.collection.immutable.Vector[Int] = Vector(5, 1, 3)
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